12th hacker series

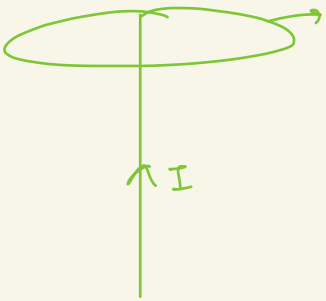
Electromagnetic waves

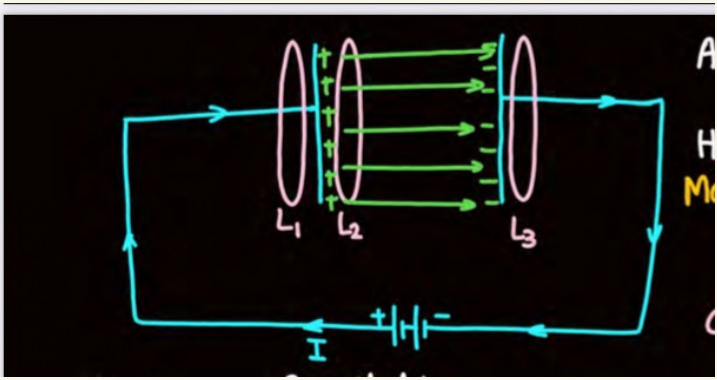
Electromagnetic waves

- we have learned that an electric current produces a magnetic field
- Also magnetic field changing with time produces an electric field
- Can an electric field changing with a time produce a magnetic field?

James clerk Maxwell (1831- 1879) argued that this was indeed the case and electric field changing with time produces a magnetic field.

- Maxwell noticed that ampere circuit law is inconsistent, namely makes non-unique predictions for the magnetic field in situations where electric current changes with time
- he showed that consistency requires an additional source of magnetic field. This is called displacement current.
- This made the laws of electricity and magnetism symmetrical





By ampere's circuital law,

$$\oint_1 \vec{B} \cdot d\vec{l} = \mu_0 I \quad \text{—— (1)}$$

$$\oint_2 \vec{B} \cdot d\vec{l} = \mu_0 (0) = 0 \quad \text{—— (2)}$$

$$\oint_3 \vec{B} \cdot d\vec{l} = \mu_0 (I) \quad \text{—— (3)}$$

As $\oint_1 \vec{B} \cdot d\vec{l} \neq \oint_2 \vec{B} \cdot d\vec{l}$

→ Hence, Ampere's circuital law is inconsistent.

- Maxwell modified this law

Current

Conduction Current

$$I_C = \frac{dq}{dt}$$

Displacement current

$$I_D = \varepsilon_0 \frac{d\Phi_E}{dt}$$

By gauss law

$$I_D = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right)$$

$$I_D = \frac{dq}{dt}$$

Modified ampere circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_{net})$$

$$= \mu_0 (I_C + I_D)$$

$$\boxed{\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_C + \varepsilon_0 \frac{d\Phi_E}{dt}\right)}$$

Now if we apply,

$$\oint_{L_1} \vec{B} \cdot d\vec{l} = \mu_0 (I_c) = \mu_0 I$$

$$\oint_{L_2} \vec{B} \cdot d\vec{l} = \mu_0 (I_D) = \mu_0 I$$

$$\oint_{L_3} \vec{B} \cdot d\vec{l} = \mu_0 (I_c) = \mu_0 I$$

As $\oint_{L_1} \vec{B} \cdot d\vec{l} = \oint_{L_2} \vec{B} \cdot d\vec{l}$

Hence, modified ampere circuit law is consistent

Displacement current

$$I_D = \varepsilon_0 \frac{d\phi_E}{dt} = \varepsilon_0 \frac{d}{dt}(EA$$

$$\boxed{I_D = \varepsilon_0 A \frac{dE}{dt}}$$

$$I_D = \varepsilon_0 A \frac{d}{dt}\left(\frac{V}{d}\right)$$

$$\boxed{I_D = \frac{\varepsilon_0 A}{d} \frac{dV}{dt} = C\frac{dV}{dt}}$$

- Maxwell formulated a set of four equations called Maxwell's equation
- With the help of these equations, he predicted the electric and magnetic field is dependent on time and space and propagate as transverse waves called electromagnetic waves
- His discovery that electromagnetic waves travel with the speed of light led him to a remarkable conclusion that light is an electromagnetic wave

Maxwell's Equations in Vacuum

$\oint \mathbf{E}\cdot d\mathbf{A} = Q/\varepsilon_0$	(Gauss's Law for electricity)
$\oint \mathbf{B}\cdot d\mathbf{A} = 0$	(Gauss's Law for magnetism)
$\oint \mathbf{E}\cdot d\mathbf{l} = \frac{-d\Phi_B}{dt}$	(Faraday's Law)
$\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$	(Ampere – Maxwell Law)

Displacement current

- displacement current is that current which comes into existence in addition to the conduction current whenever the electric field and the electric flux changes with time

$$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$

QUESTION (CBSE 2024)

In the four regions, I, II, III and IV, the electric fields are described as:
Region I : $E_x = E_0 \sin(kz - \omega t)$
Region II: $E_x = E_0$
Region III: $E_x = E_0 \sin kz$
Region IV: $E_x = E_0 \cos kz$
The displacement current will exist in the region:

A) I
B) IV
C) II
D) III

QUESTION (CBSE 2024) [1 Mark]

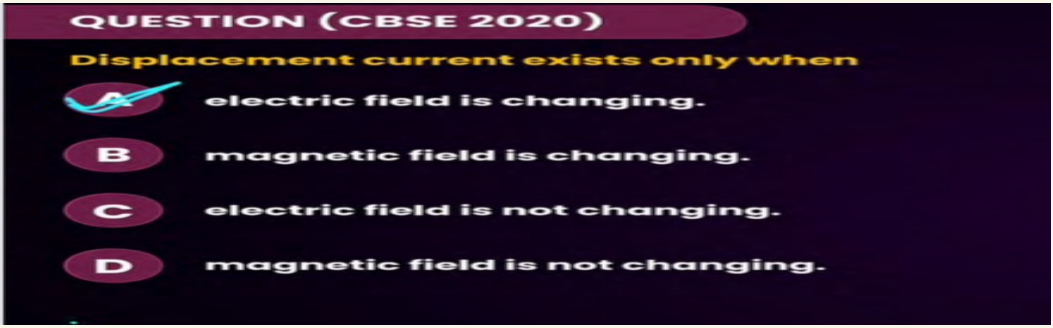
The electric and magnetic fields of electromagnetic waves are:

A) In the same phase and perpendicular to each other.
B) In the same phase and not perpendicular to each other.
C) Not in the same phase but are perpendicular to each other.
D) Neither in the same phase nor perpendicular to each other.

$\vec{E} = E_0 \sin(\underbrace{\omega t - kx}_{\text{Phase}})\,\hat{j}$

$\vec{B} = B_0 \sin(\underbrace{\omega t - kx}_{\text{Phase}})\,\hat{k}$





Maxwell modification of ampere's law

- According to Maxwell. The source of a magnetic field is not just the conduction current due to the flowing charges, but also the time varying electric field.
- Hence the total current is the sum of the conduction current I_c and displacement current I_d
- Hence the modified form of the ampere's law is

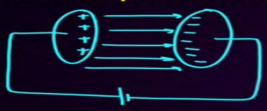
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[I_c + \varepsilon_0 \frac{d\phi_E}{dt} \right]$$

Important properties of displacement current

- displacement current exist whenever there is a change in electric flux. Unlike conduction current, it does not exist under steady conditions.
- Together with the conduction, displacement current satisfies the property of continuity

QUESTION

A parallel plate capacitor has circular plates, each of radius $5.0\ \text{cm}$. It is being charged so that electric field in the gap between its plates rises steadily at the rate of $10^{12}\ Vm^{-1}s^{-1}$. What is the displacement current?



$$I_D = \varepsilon_0 \frac{d\phi_E}{dt}$$
$$= \varepsilon_0 \frac{d}{dt}(EA)$$
$$= \varepsilon_0 A \frac{dE}{dt}$$
$$I_D = \varepsilon_0 \pi r^2 \frac{dE}{dt}$$
$$= 8.85 \times 10^{-12} \pi \left(\frac{5}{100}\right)^2 \times 10^{12}$$
$$= 0.07\ A.$$

QUESTION

The voltage between the plates of a parallel-plate capacitor of capacitance $1.0\ \mu\text{F}$ is changing at the rate of $5\ Vs^{-1}$. What is the displacement current in the capacitor?

$$I_D = C \frac{dV}{dt}$$
$$= 1 \times 10^{-6} \times 5$$
$$= 5 \times 10^{-6}\ A.$$

QUESTION (CBSE 2023)

An electromagnetic wave is produced by a charge [1 Mark]

A. moving with a constant velocity

B. moving with a constant speed parallel to a magnetic field

C. moving with an acceleration

D. at rest

Maxwell equation

1. $\oint \mathbf{E}\cdot d\mathbf{A} = Q/\varepsilon_0$ (Gauss's Law for electricity)
2. $\oint \mathbf{B}\cdot d\mathbf{A} = 0$ (Gauss's Law for magnetism)
3. $\oint \mathbf{E}\cdot d\mathbf{l} = \frac{-d\Phi_B}{dt}$ (Faraday's Law)
4. $\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$ (Ampere – Maxwell Law)

QUESTION

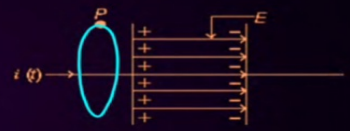
Maxwell's work led to the conclusion that light is an electromagnetic wave. His work thus unified

A electricity and magnetism
B magnetism and light
C electricity and light
D electricity, magnetism and light

QUESTION

In the given figure, for which configuration the magnetic field (say at point P) outside the plates of the capacitor is to be same everywhere?

A Plane circular loop perpendicular to wire
B Plane circular loop parallel to wire
C Pot like surface with bottom between the plates
D Tiffin box like shape (without the lid)



QUESTION

The charge on a parallel-plate capacitor varies as $q = q_0 \cos 2\pi\nu t$. The plates are very large and close together (area $= A$, separation $= d$). Find the displacement current through the capacitor.

A $-\sin 2\pi\nu t\,(2\pi\nu)$
B $q_0 \cos 2\pi\nu t$
C $2\pi q_0 \nu \sin(2\pi\nu t)$
D $-2\pi q_0 \nu \sin(2\pi\nu t)$

$I_D = \frac{dq}{dt}$
$= q_0[-\sin 2\pi\nu t]\,2\pi\nu$

QUESTION

What is an instantaneous rate of change of voltage for displacement current of 10A, in the space between the parallel plate capacitor of $1\mu F$?

A $10^5\ Vs^{-1}$
B $10^6\ Vs^{-1}$
C $10^{-6}\ Vs^{-1}$
D $10^7\ Vs^{-1}$

$I_D = C\frac{dV}{dt}$
$10 = 10^{-6}\frac{dV}{dt}$
$\frac{dV}{dt} = 10^7\ V/s.$

QUESTION

Maxwell's equations regarding electromagnetic wave describe the fundamental laws of

Assam CEE-2020

(a) Electricity only
(b) Magnetism only
(c) Mechanics only
(d) Both of electricity and magnetism

QUESTION

The source of displacement current is _____

GUJCET 2020

(a) Static Electric Field
(b) Changing Electric Field
(c) Changing Magnetic Field
(d) Static Magnetic Field

QUESTION

The concept of displacement current solves an ambiguity in

TS-EAMCET-03.05.2019, Shift-2

(a) Gauss's law
(b) Faraday's law
(c) Ampere's law
(d) Coulomb's law

QUESTION (HW)

The charging current for a capacitor at any instant is 0.78 A . The displacement current across the capacitor plates at that instant is

AMU-2017

(a) $\frac{0.78}{\epsilon_0} A$
(b) $0.78\mu_0 A$
(c) $\frac{0.78}{2} A$
(d) 0.78 A

QUESTION

An electric charge in uniform motion produces

CG PET- 2014

(a) Only electric field
(b) Only magnetic field
(c) Both electric and magnetic field
(d) Neither electric nor magnetic field

QUESTION

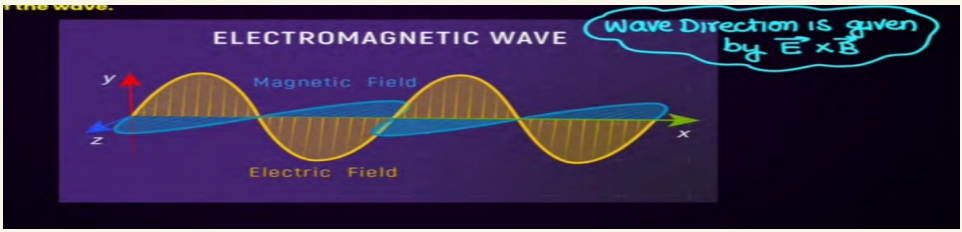
Which of the law representing Maxwell's third equation?

J&K-CET-2014

(a) Gauss's law for electricity
(b) Ampere-Maxwell law
(c) Gauss's law of magnetism
(d) Faraday's law

Maxwell's prediction of electromagnetic waves

- in 1865, Maxwell theoretically predicted the existence of electromagnetic waves
- According to faradays law of electromagnetic induction: a time varying magnetic field is a source of changing electric field
- on the basis of his theoretical studies, Maxwell argued that a time worrying electric field is a source of changing magnetic field
- This means that the change in either field, electric or magnetic produces the other field
- Maxwell further showed that these variations in electric and magnetic fields occur, occur in mutually perpendicular directions and have wave like properties
- he was led to the idea that a wave of electric and magnetic fields both varying with space and Time should exist while providing the source of the other
- Such a wave is called electromagnetic wave and it indeed exist
- an electromagnetic wave is a wave radiated by an accelerated charge, and which propagates through space as coupled, electric and magnetic field is oscillating perpendicular to each other and to the direction of propagation of waves



Plane progressive wave

$y = A \sin(kx - \omega t)$ → + x direction

- y: displacement of particle
- A: amplitude
- $(kx - \omega t)$: position of wave

$\vec{E} = E_0 \sin(kx - \omega t) \cdot \hat{j}$

- E_0: max. value of EF

$k = \frac{2\pi}{\lambda}$ → wave no.

$\omega = \frac{2\pi}{T} = 2\pi f$ → angular velocity

Mathematical representation of electromagnetic waves

$\vec{E} = E_y \hat{j} = E_0 \sin(kx - \omega t)\hat{j}$

$\vec{B} = B_z \hat{k} = B_0 \sin(kx - \omega t)\hat{i}$

Wave direction $\rightarrow \vec{E} \times \vec{B}$

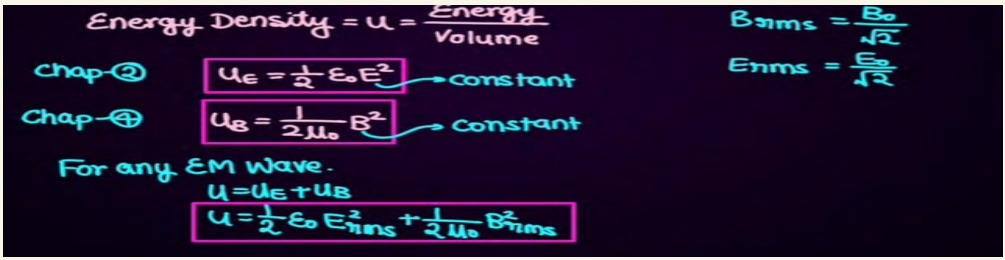
where $\boxed{\frac{E_0}{B_0} = c}$

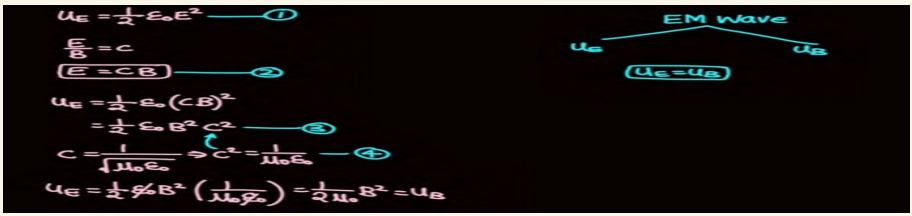
- maxwell also showed that the speed of electromagnetic wave depends on the permeability and permittivity of the medium through which it travels. The speed of electromagnetic wave in free space is given by.

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}}$$

Energy density of an electromagnetic wave

- electromagnetic wave carry energy as they travel through the space, and this energy shared equally by the electric and magnetic field. Energy density of an electromagnetic wave is the energy in unit volume of the space through which the wave travels.





- if free space the energy density of the static field E is

$$u_E = \frac{1}{2}\varepsilon_0 E^2$$

- again, in free space the energy density of the static magnetic field is

$$u_B = \frac{1}{2\mu_0} B^2$$

- the total energy density of the static electric and magnetic fields will be

$$u = u_E + u_B = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2$$

- but in an electromagnetic wave, both E and B fields, vary sinusoidally in space and Time
- the average energy density of an electromagnetic wave can be obtained by replacing E and B by their rms value in the above equation
- thus $\boxed{u = \frac{1}{2}\varepsilon_0 E_{rms}^2 + \frac{1}{2\mu_0} B_{rms}^2}$

Properties of electromagnetic waves

- all electromagnetic wave travel in free space with the same speed

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}} = 3 \times 10 \text{ m/s}$$

in a material medium, the electromagnetic waves travel with the same speed

$$\boxed{v = \frac{1}{\sqrt{\mu\varepsilon}}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{c}{n}$$

where x is the refractive index of the median

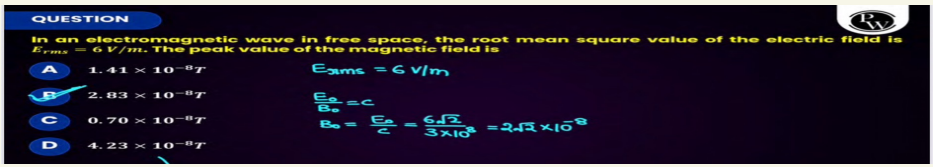
- the amplitude ratio of the electric and magnetic field is $\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$
- electromagnetic waves carry energy as they travel through the space and this energy is shared equally by the electric and magnetic field. The average energy density of an electromagnetic wave is.

$$u = u_E + u_B = \frac{1}{2}\left[\varepsilon_0 E_0^2 + \frac{B_0^2}{\mu_0}\right]$$

- electromagnetic waves transport linear momentum as they travel through space $p = \frac{u}{c}$
- electricomagnetic waves are not deflected by electric and magnetic field
- Electromagnetic waves obey the principle of superimposition, they show the properties of reflection, refraction, interference, diffraction, and polarisation
- The electric field of an electromagnetic wave is responsible for its optical effect because $E_0 \gg B_0$









QUESTION

Electromagnetic waves travel in a medium at a speed of $2.0 \times 10^8\ ms^{-1}$. The relative permeability of the medium is 1.0. Find the relative permittivity.

$V = 2\times10^8$ m/s

$\mu_m = 1$

$V = \frac{c}{\sqrt{\mu_m \varepsilon_m}}$

$2\times10^8 = \frac{3\times10^8}{\sqrt{1\,\varepsilon_m}}$

$\sqrt{\varepsilon_m} = \frac{3}{2} \Rightarrow \boxed{\varepsilon_m = \frac{9}{4}}$

QUESTION

Electromagnetic waves travel in a medium at a speed of $2.0 \times 10^8\ ms^{-1}$. The relative permeability of the medium is 1.0. Find the relative permittivity.

Solution.
Speed of an e.m. wave in a medium is given by

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_r\mu_0\varepsilon_r\varepsilon_0}}$$
$$= \frac{1}{\sqrt{\mu_0\varepsilon_0}}\frac{1}{\sqrt{\mu_r\varepsilon_r}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$
$$v^2 = \frac{c^2}{\mu_r\varepsilon_r}$$

Hence relative permittivity,

$$\varepsilon_r = \frac{c^2}{\mu_r v^2} = \frac{(3\times10^8)^2}{1.0\times(2\times10^8)^2} = 2.25.$$

QUESTION

A plane electromagnetic wave of frequency 25MHz travels in free space along the x-direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}\ Vm^{-1}$. What is $\vec{B}$ at this point? [NCERT]

$\vec{E} = 6.3\ \hat{j}$ V/m

$\frac{E_0}{B_0} = c$

$B_0 = \frac{E_0}{c} = \frac{6.3}{3\times10^8} = 2.1\times10^{-8}$ T

$\vec{B} = 2.1\times10^{-8}\ \hat{k}$ T

$E_{rms} = \frac{E_0}{\sqrt{2}}$

$E_0 = \sqrt{2}\,E_{rms}$

QUESTION

A plane electromagnetic wave of frequency 25MHz travels in free space along the x-direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}\ Vm^{-1}$. What is $\vec{B}$ at this point? [NCERT]

Solution.
Magnitude of $\vec{E}$, $E = 6.3\ Vm^{-1}$
Magnitude of $\vec{B}$,

$$B = \frac{E}{c} = \frac{6.3\ Vm^{-1}}{3\times10^8\ ms^{-1}} = 2.18\times10^{-8}\ T$$

We note that $\vec{E}$ is along y-direction and the wave propagates along x-direction. **Now an e.m. wave propagates in the direction of the vector $\vec{E}\times\vec{B}$.** So this vector must point along x-direction. As

$$(+\hat{j})\times(+\hat{k}) = +\hat{i}$$

Clearly, vector $\vec{B}$ is along z-direction. We can write

$$\vec{B} = 2.18\times10^{-8}\hat{k}\ T.$$

QUESTION

Wave → −x
MF → y
$\vec{E}\times\vec{B} = -\hat{i}$

The magnetic field in a plane electromagnetic wave is given by
$\vec{B} = 2\times10^{-7}\sin(0.5\times10^3 x + 1.5\times10^{11}t)T$.
(a) What is the wavelength and frequency of the wave?
(b) Write an expression for the electric field. [NCERT]

(a) On comparing,
$B = B_0\sin(Kx+\omega t)$
$K = 0.5\times10^3$
$\frac{2\pi}{\lambda} = 0.5\times10^3$
$\lambda = \frac{2\pi}{0.5\times10^3} = 4\pi\times10^{-3}$ m

$\omega = 1.5\times10^{11}$
$2\pi f = 1.5\times10^{11}$
$f = \frac{1.5}{2\pi}\times10^{11}$
$f = 0.24\times10^{11}$ Hz

(b) $\frac{E_0}{B_0} = c$
$E_0 = cB_0$
$= 3\times10^8 \times 2\times10^{-7}$
$= 60$ V/m
$\vec{E} = 60\sin(0.5\times10^3 x + 1.5\times10^{11}t)\hat{k}$ V/m

QUESTION

The magnetic field in a plane electromagnetic wave is given by
$B_y = 2\times10^{-7}\sin(0.5\times10^3 x + 1.5\times10^{11}t)T$.
(a) What is the wavelength and frequency of the wave?
(b) Write an expression for the electric field.

Solution.
Wavelength,

$$\lambda = \frac{2\pi}{0.5\times10^3} = \frac{2\times3.14}{0.5\times10^3}$$
$$= 1.26\times10^{-2}\ m = 1.26\ cm.$$

Also, $2\pi\nu = 1.5\times10^{11}$

$$\therefore\ \nu = \frac{1.5\times10^{11}}{2\pi} = \frac{1.5\times10^{11}}{2\times3.14}$$
$$= 23.9\times10^9\ Hz = 23.9 GHz.$$

(b) $B_0 = 2\times10^{-7}$ T

$$\therefore E_0 = cB_0 = 3\times10^8\times2\times10^{-7} = 60 Vm^{-1}$$
$$E_y = 60\sin(0.5\times10^3 x + 1.5\times10^{11}t)Vm^{-1}$$

QUESTION

In a plane electromagnetic wave travelling in free space, the electric field component oscillated sinusoidally at a frequency of $2.0 \times 10^{10} Hz$ and amplitude $48 Vm^{-1}$. Then the amplitude of oscillating magnetic field is : (Speed of light in free space $= 3 \times 10^8 ms^{-1}$)

NEET (UG)-07.05.2023

$\frac{E_0}{B_0} = c$

$B_0 = \frac{E_0}{c} = \frac{48}{3 \times 10^8} = 16 \times 10^{-8}\ T$

(a) $1.6 \times 10^{-8} T$

(b) $1.6 \times 10^{-7} T$

(c) $1.6 \times 10^{-6} T$

(d) $1.6 \times 10^{-9} T$

QUESTION

Choose the correct answer from the options given below :

JEE Main-25.01.2023, Shift-I

(a) A-1, B-2, C-3, D-4

(b) A-4, B-1, C-2, D-3

(c) A-3, B-4, C-1, D-2

(d) A-2, B-3, C-4, D-1

	List-I		List-II
A.	Gauss's Law in Electrostatics	1.	$\oint \vec{E}.d\vec{l} = -\frac{d\phi_B}{dt}$
B.	Faraday's law	2.	$\oint \vec{B}.d\vec{A} = 0$
C.	Gauss's Law in Magnetism	3.	$\oint \vec{B}.d\vec{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$
D.	Ampere-Maxwell Law	4.	$\oint \vec{E}.\vec{ds} = \frac{q}{\varepsilon_0}$

QUESTION

In a plane EM wave the electric field oscillates sinusoidally at a frequency of 30 MHz and amplitude 150 V/m. Identify the correct expression of $\vec{B}$ assuming the wave is propagating along x-axis and is oscillating along y-axis.

TS EAMCET 20.07.2022, Shift-II

$\omega = 2\pi (30 \times 10^6)$

$= 60\pi \times 10^6$

(a) $5 \times 10^{-7} \sin\left[\frac{x}{3} - 6 \times 10^{+7} t\right] \hat{z}\ T$

(b) $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{5} - 6 \times 10^{+7} t\right)\right] \hat{z}\ T$

(c) $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{10} - \times 10^{+7} t\right)\right] \hat{z}\ T$

(d) $5 \times 10^{-7} \sin\left[\pi\left(\frac{2x}{5} - \times 10^{+8} t\right)\right] \hat{z}\ T$

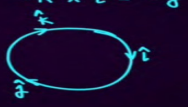
QUESTION

In an electromagnetic wave, at an instant and at a particular position, the electric field is along the negative z-axis and magnetic field is along the positive x-axis. Then the direction of propagation of electromagnetic wave is:

JEE Main-13.04.2023, Shift-II

$\vec{E} \times \vec{B}$

$-\hat{k} \times \hat{i} = -\hat{j}$



(a) at 45° angle from positive y -axis

(b) negative y-axis

(c) positive z-axis

(d) positive y-axis

QUESTION

Statement I: A time varying electric field is a source of changing magnetic field and vice-versa. Thus a disturbance in electric or magnetic field creates EM waves.

Statement II: In a material medium, the EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

$V = \frac{c}{\sqrt{\mu_m \varepsilon_m}}$

JEE Main-27.06.2022, Shift-II

(a) Both statement I and statement II are true.

(b) Both statement I and statement II are false.

(c) Statement I is correct but statement II is false.

(d) Statement I is incorrect but statement II is true.

QUESTION

Light wave traveling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct) Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 ms^{-1}$)

JEE Main-25.07.2022, Shift-II

$B_0 = \frac{E_0}{c} = \frac{540}{3 \times 10^8} = 180 \times 10^{-8}$

$= 18 \times 10^{-7}\ T.$

(a) $18 \times 10^{-7} T$

(b) $54 \times 10^{-7} T$

(c) $54 \times 10^{-8} T$

(d) $18 \times 10^{-8} T$

QUESTION

Which of the following has/have zero average value in a plane electromagnetic wave?

COMEDK 2020

(a) Both magnetic and electric fields

(b) Electric field only

(c) Magnetic field only

(d) Magnetic energy

QUESTION

In an electromagnetic wave

VITEEE–2019

(a) power is transmitted along the magnetic field

(b) power is transmitted along the electric field

(c) power is equally transferred along the electric and magnetic fields

(d) power is transmitted in a direction perpendicular to both the fields

QUESTION

For the plane electromagnetic wave given by $E = E_0 \sin(\omega t - kx)$ and $B = B_0 \sin(\omega t - kx)$, the ratio of a average electric energy density to average magnetic density is

JEE Main–06.04.2023, Shift–I

$\frac{u_E}{u_B} = 1$

(a) 1

(b) 1/2

(c) 2

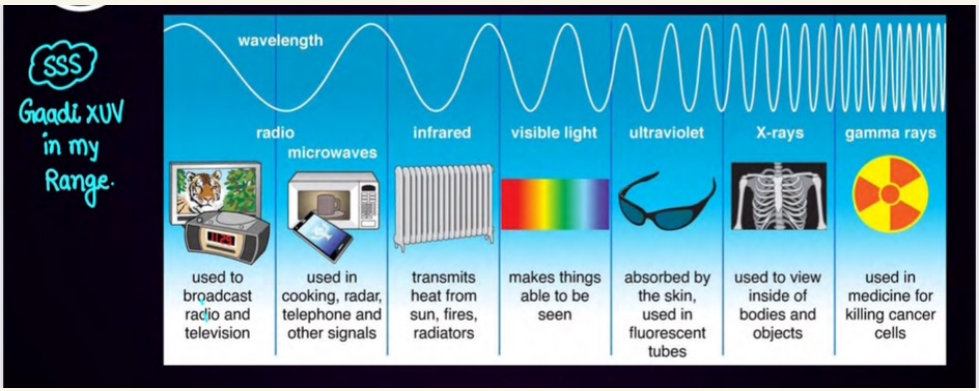
(d) 4

Electromagnetic spectrum

- the orderly distribution of the electromagnetic waves, according to their wavelength of frequency into distinct groups having widely differing properties is called electromagnetic spectrum
- The main part of the electromagnetic spectrum are Gama ray, x-rays, ultraviolent ray, visible light, infrared rays, microwaves, and radio waves in order of the increasing wavelength from 10^{-12} m to 10^{6} m

$\nu_{max} \rightarrow \lambda_{min} \rightarrow$ gamma rays

$\nu_{min} \rightarrow \lambda_{max} \rightarrow$ radio waves



Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer. Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	$<10^{-3}$ nm	Radioactive decay of the nucleus	-do-

not to be © MP

Radio waves

- these are the electromagnetic waves of longest wavelength and minimum frequency
- Wavelength range- 600 m to 0.1 m
- Frequency range: 500 kHz to 1000 MHz
- Source: accelerated motion of charges in conducting wires or oscillating circuits
- Properties: reflection, diffraction
- Uses: In radio and television communication systems
  In radio astronomy

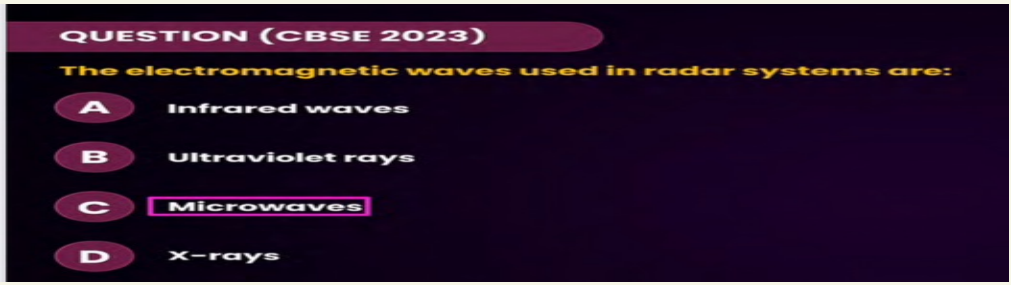
Table- Some important wireless communication bands

Frequency band	Service
$540 - 1600 kHz$	Medium wave AM band
$3 - 30 MHz$	Shortwave AM band
✓ $88 - 108 MHz$	FM broadcast
✓ $54 - 890 MHz$	TV Waves
✓ $840 - 935 MHz$	Cellular Mobile radio

Microwaves

- they are the electromagnetic waves having wavelength smaller than radio waves
- Wavelength range: .3m to 10^{-3} m
- Frequency range: 10^{9} Hz to 10^{12} Hz
- Source: osculating, currency and special vacuum tubes like magnetron and Gunn diodes

- properties: reflection, refraction, diffraction, and polarisation. Due to their shorter wavelength, they can travel as a beam in a signal.
- Uses of microwave- In radar system for aircraft navigation
  In long distance communication system via geostationary satellites
  In microwave oven



Infrared waves

- these radiation lie close to the low frequency or long wavelength of the visible spectrum
- Infrared waves produce heating effect, so they are also known as heat waves of thermal radiations
- The water molecules present in different materials. Readily absorb infrared waves, increase the thermal motions and hence heat up the material and their surroundings.
- Wavelength range : - 5×10^{-3} m to 10^{-6} m
- Frequency range : 10^{11} Hz to 5×10^{14} Hz
- Source: hot bodies and molecules
- Properties: heating effect, reflection, refraction, and propagation through fog
- Uses of infrared waves:
  → in the remote control of the TV or VCR, the keypad of which contains a small internet transmitter
  → In greenhouses to keep the plants warm
  → In Hayes photography, because interested waves are less scattered than visible light by atmospheric particles

→ Infrared lamps in the treatment of muscular complaints
→ In reading the secret writing on the ancient walls
→ Knowing the molecular structure

Visible light

- it is a very small part of the electromagnetic spectrum towards which the humans are sensitive
- The visible light emitted or reflected from bodies around us, gives the information about the world
- Wavelength range: 8×10^{-7} m to 4×10^{-7} m
- Frequency range: 4×10^{14} Hz to 7×10^{14} Hz
- Source: radiated by excited atoms in ionised gas and incandescent bodies
- Properties: reflection, refraction, interference, refraction, polarisation, photoelectric, effect, photographic action, sensation of sight
- Uses: It provides us the information of the world around
  It can cause chemical reactions
  The approximate wavelength ranges for lights of different colours are as follows

V	Violet	400 nm-420 nm
I	Indigo	420 nm-450 nm
B	Blue	450 nm-500 nm
G	Green	500 nm-570 nm
Y	Yellow	570 nm-600 nm
O	Orange	600 nm-650 nm
R	Red	650 nm-750 nm

Ultraviolet light

- this region of the electromagnetic spectrum has wavelength just shorter than visible light and can be detected just beyond the violet end of the solar spectrum
- Wavelength range: -3.5×10^{-7} m to 1.5×10^{-7} m

- frequency range: 10^{16} Hz to 10^{17} Hz
- Source: high voltage gas, discharge tubes, arcs of iron and mercury the sun
- Properties: effect on photographic plate, fluorescence, ionisation, highly energetic , tanning of the human skin
- Uses: In food preservation
  in the study of invisible writings, forged documents-and fingerprints
  in the study of molecular structure

- The ultraviolent light in large quantities have harmful effect on human being.
- But fortunately, most of the ultraviolent light coming from the sun is absorbed by the ozone layer in the atmosphere at an altitude of about 40 to 50 km

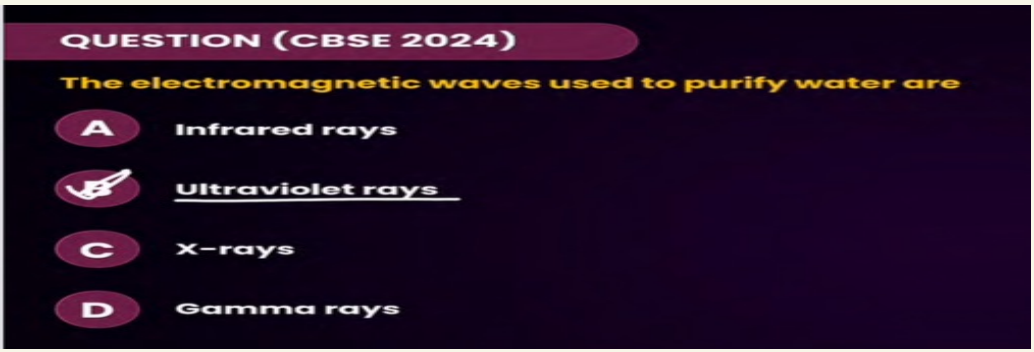
X-ray

- these electromagnetic waves have wavelength just shorter than the ultraviolent light
- as x-rays can pass through many forms of matter, so they have many useful, medical and industrial application
- Wavelength range: 100 Å to 0.1Å
- Frequency range: 10^{18} Hz to 10^{20} Hz
- Source: sudden deceleration of fast moving electrons by a metal target
- Properties: effect on photographic plate, ionisation of gases, photoelectric effect, fluorescence, more energetic than the UV rays
- Uses: In medical diagnosis because x-ray can pass through flash, but not through bones
  In the study of Crystal structure because x-ray can be reflected and deflected by crystals
  An engineering for detecting false cracks, flaws and holes in the finished metal products
  In de aath Echt departments to detect explosive, diamond gold, et cetera, in the position of smugglers
  In radiotherapy to cure, untraceable, skin, disease, diseases, and malignant growth

Gamma rays

- these are electromagnetic radiations of highest frequency range and lowest wavelength range
- These are most penetrating electromagnetic wave
- Wavelength range: 10^{-14} m to 10^{-10} m
- Frequency range : 10^{18} Hz to 10^{22} Hz
- Source: radioactive nuclei and nuclear reactions. Is a pure Gamma ray source
- Properties: effect on photographic plate, fluorescence, ionisation , diffraction , high penetrating power
- Uses: In radiotherapy for the treatment of malignant tumours
  In the manufacture of polyethylene from ethylene
  To initiate some nuclear reaction
  To preserve food stuffs for a long time, because soft, ray, and kill microorganism
  To study the structure of atomic nuclei





QUESTION

The part of the spectrum of the electromagnetic radiation used to cook food is

J&K CET- 2011

(a) ultraviolet rays

(b) cosmic rays

(c) X rays

(d) microwaves ✓

QUESTION

The crystal structure can be studied by using

COMEDK 2017
J&K CET- 2010
AIPMT- 1992

(a) UV rays

(b) X-rays ✓

(c) IR radiation

(d) Microwaves

QUESTION

Choose the electromagnetic radiation relevant to telecommunication

JCECE-2018
AMU-2013

(a) ultraviolet

(b) infrared

(c) visible

(d) microwave ✓

Ans. (d) : Microwaves have comparatively high enough frequency, than the other waves, by which it carries the information without any absorption or scattering due to any obstacle in the way. On the other hand visible waves, infrared waves and UV waves can be affected by scattering, hard objects like fog, walls and absorption respectively.

QUESTION

The range of wavelength of the visible part of the electro magnetic spectrum is

AMU-2001

(a) $8500 - 10000\text{Å}$

(b) $7000 - 8500\text{Å}$

(c) $3000 - 4000\text{Å}$

(d) $4000 - 7000\text{Å}$ ✓

QUESTION

The electromagnetic wave with shortest wavelength among the following is

NEET (Oct.) - 2020

(a) UV-rays

(b) X-rays

(c) γ-rays ✓

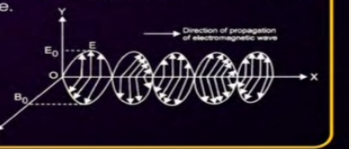
(d) microwaves

1. **Electromagnetic waves:** The oscillating or accelerated charged particle produces an oscillating electric field in space, which in turn produces an oscillating magnetic field. These oscillating electric field and magnetic field are mutually perpendicular to each other give rise to an electromagnetic wave after superimposition.
The two fields become source of each other and the wave propagates sinusoidally in a direction perpendicular to both the fields.

Note: Electromagnetic waves are not produced by (a) a neutral particle (b) a stationary charged particle (c) a charged particle moving with constant velocity.

The frequency of electromagnetic wave is equal to the frequency of accelerated charged particle.

2. **Transverse nature of electromagnetic wave :**
The variation of electric field ($\vec{E}$) and magnetic field ($\vec{B}$) are mutually perpendicular to each other as well as to the direction of propagation of the wave, i.e., the electromagnetic wave are transverse in nature.
Electromagnetic wave are not deflected by electric and magnetic fields.
Electric and magnetic field in electromagnetic wave are in same phase.
A plane electromagnetic wave propagating along x-direction is shown in figure.



The electric and magnetic fields are represented respectively as follows:
$E_y = E_0 \sin(kx - \omega t)$; $B_z = B_0 \sin(kx - \omega t)$,
where $k = \frac{2\pi}{\lambda}$, is the magnitude of the propagation vector $\vec{k}$.
The speed of electromagnetic wave
$c = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$
$= 3 \times 10^8\ ms^{-1}$

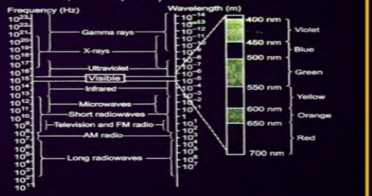
Note:
(a) In a medium, the speed of electromagnetic wave is given by
$v = \frac{1}{\sqrt{\mu\varepsilon}}$
(b) Refractive index of the medium is given by
$\mu = \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0 \varepsilon_0}} = \sqrt{\mu_r \varepsilon_r}$

3. **Energy density of electromagnetic waves:**
The energy per unit volume of the space through which it travelled. Electromagnetic wave carry energy as they travel through the space. This energy is shared by the electric and magnetic fields.
$U_B = \frac{B^2}{2\mu_0}$ [U_B = energy of magnetic field]

4. **Intensity of electromagnetic waves:**
The energy crossing per unit area per unit time perpendicular to the direction of propagation of the wave.
$$\text{Intensity} = \frac{\text{Energy}}{\text{Area} \times \text{Time}}$$
$$I = \frac{1}{\mu_0} B_{rms}^2 c = \frac{1}{2} \varepsilon_0 E_0^2$$

5. **Important characteristics of electromagnetic waves:**
(i) The energy in electromagnetic wave is divided on average equally between electric and magnetic field.
(ii) Energy associated with an electromagnetic wave is $U = \frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}$
(iii) Electromagnetic waves are transverse in nature.
(iv) Electromagnetic waves also carry energy and momentum. Linear momentum delivered to the surface by an electromagnetic wave is $p = \frac{U}{c}$, where U is the total energy transmitted by electromagnetic wave and c is the speed of electromagnetic wave.
(v) Electromagnetic wave also exerts pressure called radiation pressure.
(vi) An accelerated charged particle is a basic source of electromagnetic waves.
(vii) Electromagnetic waves can be reflected, refracted and diffracted.

6. **Electromagnetic spectrum:** The whole range of frequencies/wavelength of the electromagnetic waves arranged in ascending or descending order is known as the electromangetic spectrum. The wavelength range varies from 10^{-12} m to 10^4 m i.e., from γ-rays to radiowaves.



The electromagnetic spectrum with common names for various part of it.

Different types of electromagnetic waves

Type	Wavelength range	Frequency range (Hz)	Production	Detection
Radiowave	> 0.1 m	10^4 to 10^9	Rapid acceleration and deceleration of electrons in aerials.	Receiver's aerials
Microwave	0.1 m to 1 mm	10^9 to 10^{11}	Klystron valve or magnetron valve.	Point contact diodes
Infrared wave	1 mm to 700 nm	3×10^{11} to 4×10^{14}	Vibration of atoms and molecules.	Thermopile, Bolometer, infrared photographic film
Light	700 nm to 400 nm	4×10^{14} to 8×10^{14}	Electrons in atoms emit light when they move from one energy level to a lower epergy level.	The eye, photocells, photographic film
Ultraviolet rays	400 nm to 1 nm	8×10^{14} to 8×10^{16}	Inner shell electrons in atoms moving from one energy level to a lower level.	Photocells, photographic film
X-rays	1 nm to 10^{-3} nm	1×10^{16} to 3×10^{21}	X-ray tubes or inner shell electrons.	Photographic film, Geiger tubes, ionisation chamber
γ-rays	< 10^{-3} nm	5×10^{18} to 5×10^{22}	Radioactive decay of the nucleus.	Photographic film, ionisation chamber

Note: Trick to learn order of increasing frequency (decreasing wavelength). Rajesh Mahajan is In Visible Under X-ray eye Game.

(a) R → Radiowave
(b) M → Microwave
(c) I → Infrared waves
(d) V → Visible rays
(e) U → Ultraviolet rays
(f) X → X-rays
(g) G Gamma rays (γ)

7. Uses of electromagnetic waves:

(a) Radiowaves:
(i) In astronomical field.
(ii) In radio and TV communication.

(b) Microwaves:
(i) In RADAR communication.
(ii) For atomic and molecular research.
(iii) For cooking purpose.
(iv) For weather forecasting.
(v) In long distance communication system via geostationary satellites.

(c) Infrared waves:
(i) In remote control.
(ii) In treatment of muscular complaints.
(iii) In greenhouse to maintain temperature suitable for plants.
(iv) In knowing molecular structure.

(d) Visible rays:
(i) To see things.
(ii) Great source of energy for human life.
(iii) Useful in photography.

(e) Ultraviolet rays:
(i) To study molecular structure.
(ii) To kill germs in minerals.
(iii) To preservation in food.
(iv) Used in burglar alarm.
(v) Used in detecting the invisible writings, finger prints.
(vi) To sterlizing the surgical instruments.

(f) X-rays:
(i) Used in radio therapy to care cancer and tumors.
(ii) In detecting faults, cracks etc. in metal product.
(iii) In medical diagnosis and cure malignant growth and untraceable skin diseases as they pass through the muscles not through the bones.

(g) Gamma rays:
(i) In radio therapy for treatment of tumour and cancer.
(ii) For food preservation.
(iii) To study about the structure of atomic nuclei.

8. Ampere's Maxwell Circuital law:

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 I_C + \varepsilon_0 \mu_0 \frac{d\phi_E}{dt}$$

where B is the magnetic field, dl is small element of length l, I_C is conducting current and ϕ_E is the electric flux.

9. Displacement current:

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 (I + I_d)$$

I_d = Displacement current

$$I_d = \varepsilon_0 \left(\frac{d\phi_E}{dt}\right)$$

FORMULAS

$E = E_0 \sin(\omega t - Kx)$

$B = B_0 \sin(\omega t - Kx)$

$\omega = \frac{2\pi}{T} \quad K = \frac{2\pi}{\lambda}$

$c = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

$V = \frac{c}{\sqrt{K \mu_r}}$

EM WAVES

Energy Density

$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2$ (Static)

For Sinusoidal wave

$u = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4\mu_0} B_0^2$

$u = \varepsilon_0 E_{rms}^2 = \frac{B_{rms}^2}{\mu_0}$

$u_E = u_B$

Intensity of wave

$I = \frac{1}{2}\varepsilon_0 E_0^2 c = \varepsilon_0 E_{rms}^2 c$

$I = \frac{1}{2\mu_0} B_0^2 c = \frac{B_{rms}^2 c}{\mu_0}$

Radiation Pressure

$P_r = \frac{I}{c}$

Displacement Current

$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$

$\oint \vec{B} \cdot \vec{dl} = \mu_0 (I_c + I_d)$

Modified Ampere's Law

E M Waves

Displacement Current

$I_D = \varepsilon_0 \frac{d\phi_E}{dt}$

Modified Ampere's law

$\oint \vec{B} \cdot \vec{dl} = \mu_0 (I_c + I_D)$

Frequency of EM Wave

$f = \frac{1}{2\pi\sqrt{LC}}$

Electric & Mag. Field

$E = E_0 \sin(\omega t - Kx)$

$B = B_0 \sin(\omega t - Kx)$

$\omega = \frac{2\pi}{T} \quad K = \frac{2\pi}{\lambda}$

Speed of E-M Wave

$C = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

Speed of light in medium

$V = \frac{C}{\sqrt{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}} = \frac{C}{\mu}$

Energy Density

For Static Field

$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2$

For Sinusoidal Field

$u = \frac{1}{2}\varepsilon_0 E_{rms}^2 + \frac{1}{2\mu_0} B_{rms}^2$

In EM Wave,

$u_E = u_B$

Intensity of EM Wave

$I = \frac{1}{2}\varepsilon_0 E_0^2 c = \varepsilon_0 E_{rms}^2 c$

Radiation Pressure

$P_R = \frac{I}{c}$

Electromagnetic Spectrum

① Gamma Ray → λ_{min} f_{max}
② X-Ray
③ U.V. Ray
④ Visible light
⑤ I.R. light
⑥ Microwaves
⑦ Radio waves → λ_{max} f_{min}

Visible Spectrum

① Violet → λ_{min}
② Indigo
③ Blue
④ Green
⑤ Yellow
⑥ Orange
⑦ Red → λ_{max}

Radiowaves	10^4 to 10^9	> 0.1 m	Rapid acceleration and deaccelerations of electrons in aerials.	Receivers aerials.	Different wavelengths find specialised uses in radio communication.
Microwaves	10^9 to 10^{12}	0.1 m to 1 mm	Klystron valve or magnetron valve.	Point contact diodes.	(a) Radar communication. (b) Analysis of fine details of molecular and atomic structure. (c) Since $\lambda = 3 \times 10^{-2}$ m, useful for demonstration of all wave properties on macroscopic scale.
Infrared	10^{11} to 5×10^{14}	1 mm to 700 nm	Vibration of atoms and molecules.	Thermopiles Bolometer Infrared photographic film.	(a) Useful for elucidating molecular structure. (b) Less scattered than visible light by atmospheric particles–useful for haze photography.
Visible light	4×10^{14} to 7×10^{14}	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level.	Human eye Photocells Photographic film.	(a) Detected by stimulating nerve endings of human retina. (b) Can cause chemical reaction.
Ultraviolet	10^{16} to 10^{17}	400 nm to 1 nm	Inner shell electrons in atoms moving from one energy level to a lower level.	Photocells Photographic film.	(a) Absorbed by glass (b) Can cause many chemical reactions, *e.g.*, the tanning of the human skin. (c) Ionize atoms in atmosphere, resulting in the ionosphere.
X-rays	10^{16} to 10^{19}	1 nm to 10^{-3} nm	X-ray tubes or inner shell electrons.	Photographic film, Geiger tubes, Ionization chamber.	(a) Penetrate matter (*e.g.*, radiography) (b) Ionize gases (c) Cause fluorescence (d) Cause photoelectric emission from metals. (e) Reflected and diffracted by crystals enabling ionic lattice spacing and N_A (or wavelength) to be measured.
Gamma rays	10^{18} to 10^{22}	< 10^{-3} nm	Radioactive decay of the nucleus.	Photographic film, Geiger tubes, Ionization chamber.	Similar to X-rays.

Thank you